

Fine structural effects of L-ascorbic acid on buccal epithelium¹

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ABSTRACT Three groups of guinea pigs were maintained on separate diets. Group I received no ascorbic acid, group II received a dose just adequate for the prevention of scurvy, and group III received a high dose of 100 times the minimum. A portion of the buccal epithelium of these animals was excised after 20 days for the scorbutic group and 55 days for groups II and III. Tissues were then prepared for electron microscopy. Measurements were taken and analyzed statistically for differences in plaque-to-plaque widths of desmosomes. Those in group I were smaller than those in the other two groups. Tonofilaments appeared to be less tensile and/or less frequently attached to desmosomes in group I than in groups II and III. These two observations indicate that ascorbic acid may have a role in the maintenance of the integrity of the tonofilaments and desmosomes in buccal epithelial cells. It is concluded that, for the maintenance of optimum physical integrity of buccal epithelial cells, ascorbic acid may be necessary in larger doses than the minimum required for the prevention of scurvy. It is proposed that doses up to 100 times the minimum may be beneficial. *Am. J. Clin. Nutr.* 27: 719–727, 1974.

Linus Pauling (1) has recently stated that ascorbic acid in large quantities is both beneficial and nontoxic to humans. This has prompted reactions of concern and alarm from some medical scientists (2, 3). Data for further evaluation of these two points of view could reasonably be provided by an electron microscopic approach, but to our knowledge no one has reported on a study of the fine structural effects of varying doses of this compound. This study was directed toward elucidating some of the ultrastructural effects of vitamin C on epithelial cells in the buccal cavity of the guinea pig. Previous studies have shown that the physiological requirements and disposition of vitamin C are similar in humans and guinea pigs, with the exception that the amount of expired CO₂ resulting from ascorbic acid breakdown is lower in man (4, 5). Products of ascorbic acid metabolism excreted in the urine are the same in both species (5). For these reasons, a study of vitamin C in the guinea pig is highly applicable to theories regarding the effects of this vitamin in man. Few studies (6) have been conducted on the effects of large amounts of this vitamin and more information needs to be accumulated that will provide a more solid basis on which to prescribe "safe" and "useful" amounts of ascorbic acid in therapy.

Vitamin C is necessary for the production of collagen (7) and for the formation of the basement lamina (8). Both of these structures are considered important in the attachment of epithelial cells to the underlying lamina propria (connective tissue); thus, they presumably aid in keeping the two tissues bound together as functional units lining the interior and exterior surfaces of the body. If these extracellular products are not synthesized and maintained, the tissues of which they are a part will lose physical integrity. This loss of integrity may, in turn, result in increased susceptibility of these and deeper cells to infection. The correlation that has been shown by some investigators (9) between the ingestion of large amounts of vitamin C and a low incidence of infection could then be the result of increased tissue integrity. This study is designed to compare some of the parameters which may affect tissue integrity in three groups of animals, one without vitamin C in the diet, one on a

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"normal" nonscorbutigenic dose (control group), and a third group receiving a high dose equivalent to that recommended by Pauling (1). It involves an examination of the ultrastructure of the basement lamina, desmosomes, hemidesmosomes, and tonofilaments of germinal epithelial cells in the buccal cavity.

Materials and methods

The experimental animals were 15 guinea pigs (*Cavia* sp.) varying in weight from 200 to 500 g. These animals were divided into three groups of five each; three of the animals in each group were male and two were female. Group I was the scorbutic group (no vitamin C), group II the control, and group III the high-dose group.

An ascorbic acid-free diet for guinea pigs was obtained from Nutritional Biochemicals Corporation in Cleveland, Ohio, and fed ad libitum during the full experimental period of 55 days. Vitamin C solutions were made each day with fresh tap water and pure crystalline powdered ascorbic acid in concentrations of 1 mg/ml and 100 mg/ml. The guinea pigs were weighed weekly to determine the amount of ascorbic acid solution to be administered to each individual. Amounts were measured by pipette and given orally by eyedropper (10).

Group I (the scorbutic group) received only tap water (1.9 to 2.0 ml/animal per day) with no ascorbic acid. Group II (the control group) received 3.8 to 4.0 mg vitamin C/kg body wt per day. This is the minimum amount necessary to prevent the occurrence of the symptoms of scurvy in guinea pigs according to Stare (10) and Rosenberg (11). Group III (the high-dose group) received 380 to 400 mg/kg per day, which corresponds to 2.6 to 2.8 g/day in man and is within the dose range recommended by Pauling (1). Tap water was supplied freely to all animals between feeding periods.

Group I was maintained on the above diet for 20 days at which time definite signs of scurvy were shown by all members of the group. The remaining two groups continued on their vitamin C diets for a total of 55 days before tissues were excised.

To prevent possible alterations in cell ultrastructure by chemical anesthetics, the animals were killed by a blow on the head. Tissue samples were taken from the left cheek in the first molar region of the buccal cavity within 1 min. These were immediately placed into 3% glutaraldehyde in cacodylate buffer (0.1 M, pH 7.4) and fixed for 1 hr at room temperature. The samples were then cut into thin slices (0.2 to 0.4 mm thick) and fixed for a 2nd hr in fresh glutaraldehyde solution. The tissues were next rinsed in three 5-min changes of buffer and stored at 4°C for 2 to 8 weeks in a buffer solution containing 0.8% sucrose. Postfixation was in 1.0% osmium tetroxide in cacodylate buffer (0.1 M, pH 7.4) for 1 hr at 4°C. After three rinses in the buffer, the samples were dehydrated in 10-min changes of 30, 60, 90, and 100% (two changes) ethyl alcohol and two changes of 99.9% propylene oxide. They were embedded in Bojax epoxy mixture (12) and sectioned with glass knives on an LKB Ultratome.

These sections were mounted on uncoated stainless steel grids and stained in 2.0% aqueous uranyl acetate and Reynolds' (13) lead citrate. Sections were examined with a JEM 7A electron microscope.

One of the parameters examined in this study was the distance between the two attachment plaques of single desmosomes. These were recorded and analyzed statistically for each group. These measurements were made from the outermost (outermost from the center of the cell) border of one attachment plaque, across the cell membrane and intercellular space to the corresponding outer border of the adjoining (second) attachment plaque of the desmosome (see Fig. 3). All measurements were made with a Bausch & Lomb 7X measuring magnifier using a grid with 100 lines/cm. Summary statistics and a *t* test for significance of numerical results were performed on a Xerox Data Systems 930 computer.

Results

General appearance

At the time of tissue excision, the general macroscopic internal anatomy of the buccal cavity was noted. No obvious differences were observed between groups.

Fine structure

Measurements. For each of the three groups, measurements were made of the distance between the two attachment plaques (see Fig. 3) of 25 different desmosomes. The control group and high-dose group are nearly the same with mean values of 68.5 nm and 65.9 nm, respectively. They are both larger at the 0.05 level of significance than the scorbutic group which has a mean desmosomal width of 58.8 nm (see Table 1).

There are several special structures that occur at the base (proximal end) of the germinal epithelial cells. These were examined, measured, and compared in this study. One of these structures is the basement lamina which contains several structural layers. There is an electron transparent region (lamina lucida) immediately outside the plasma membrane at the base of the germinal cell. The second layer is an electron opaque zone (lamina densa) and is adjacent to the lamina propria (connective tissue). These two zones are of equal thickness. Hemidesmosomes are observed along the basal cell membranes. These have only one attachment plaque located along the membrane at the proximal end of the cell. The position that would ordinarily be occupied by a second attachment plaque on the membrane of a neighboring cell is taken up by the lamina

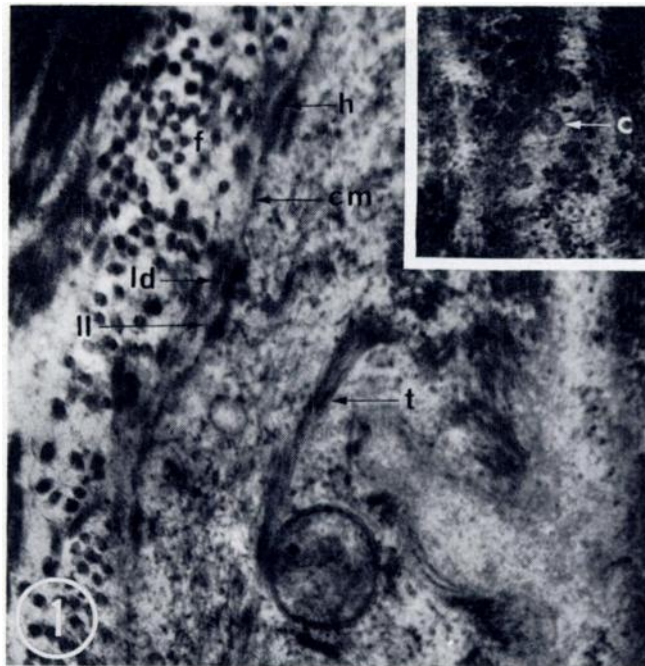


FIG. 1. Basement lamina region (proximal end of germinal cells) typical of all groups. Lamina densa (ld), lamina lucida (ll), caveolae (c) and hemidesmosomes (h) are all visible. Other prominent structures include collagen fibrils (f), tonofilaments (t) and the basal cell membrane (cm). $\times 40,000$. Inset is a higher magnification view of typical caveolae (c), representative of those seen in each group. $\times 50,000$.

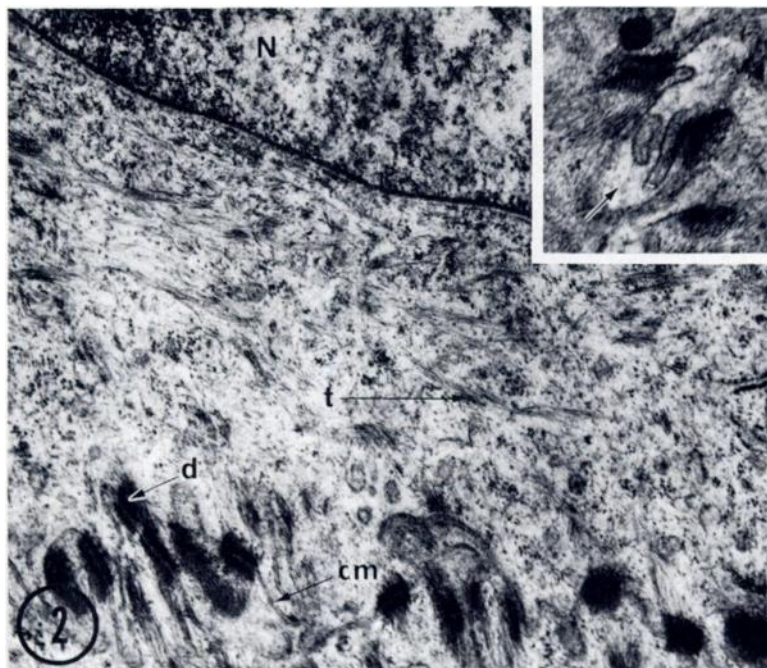


FIG. 2. Typical view of a portion of a germinal epithelial cell of group I. Note the free ribosomes and the randomly scattered bundles of tonofilaments (t). Other labeled organelles include the cell nucleus (N), desmosomes (d), and interdigitated cell membrane (cm). $\times 20,000$. Inset is an example of the enlarged intercellular spaces (arrow) often observed in this group. $\times 30,000$.



FIG. 3. High magnification of typical desmosomes in group I. Note the fragmentary appearance of the tonofilaments (t) as though they are short segments or are following a tortuous path as they extend from the attachment plaques (p) of the desmosomes. The length of the white bar between attachment plaques of one of the desmosomes demonstrates the plaque-to-plaque distance measured for statistical analysis of the desmosomal widths. $\times 75,000$.

TABLE 1
Summary statistics for desmosome widths

Ascorbic acid dose, mg/kg/day	No. measured (per group)	Mean $\pm$ SEM, nm	Mode, nm	Range, nm	SD, nm	Significance ^a
0	25	58.8 $\pm$ 1.1	60.0	46.7 66.7	5.3	Group I different from II and III
4	25	68.5 $\pm$ 1.4	66.7	60.0 80.0	6.8	
400	25	65.9 $\pm$ 0.8	66.7	60.0 73.3	4.0	

^a $P = 0.05$.

densa. In all groups, an electron opaque structure is present inside the cells and beside and parallel to each hemidesmosome attachment plaque (Fig. 1). Each of the constituents noted above is present with the same apparent structural features and dimensions in all three groups, although there is considerable variation in their size and distinctness within each group. All groups display prominent caveolae ("c" in Fig. 1), which are located only along the basal membranes of basal epithelial cells. These are microscopic vesicles, pinocytotic-like in appearance, and with diameters of approximately 50 to 150 nm.

General appearance of organelles. The germinal cell cytoplasm of the scorbutic animals contains large numbers of free ribosomes and tonofilaments randomly scattered in small bundles (Fig. 2). The tonofilaments that extend from the attachment plaques of desmosomes

most often appear as short fragments, though there are occasional regions where parallel "bundles" appear (Fig. 3). Intercellular spaces are often expanded, compared with other groups (Fig. 2, inset).

Buccal cells from organisms of the control group (II) have a marked concentration of microgranules (Fig. 4). These are electron dense bodies distributed throughout the cytoplasm of the epithelial cells. Most of them appear as spherical organelles and range in size from approximately 50 to 250 nm in diameter. They do not appear to be bound by membranes (see Fig. 4), although it is possible that a limiting membrane could be masked by their opacity. The germinal epithelium of this group contains many more of these granules than the other two groups; in fact, microgranules are rarely visible in tissues from groups I and III. A view of typical desmosomes as seen in most

sections from group II is shown in Fig. 5. Although not sharply defined, the tonofilaments still often appear as short segments, the same as has been noted previously from group I. Some polyribosomes are visible in group II cells but they are not present to any marked degree.

Polyribosomes are often seen in group III cells (Fig. 6, inset). Tonofilaments are more densely distributed throughout the cells of this group than those of the other two. After observing many sections from many tissue blocks we are impressed by the large number of tonofilaments seen extending from attachment plaques of the desmosomes for longer distances in the plane of sectioning than has been noted for groups I and II (compare Figs. 2, 4, 6 and 7). These results are notably uniform throughout the specimens examined.

Both smooth and rough endoplasmic reticulum are present only in short and infrequent segments in each group. For a summary of the preceding data see Table 2.

Discussion

Our ultimate interest in this study was to help to elucidate further the role of ascorbic acid in man. As this research was not conducted at a medical facility, human volunteers were not available. The guinea pig was considered a relevant substitute, as previous investigations have shown that experimental results concerning vitamin C in the guinea pig correspond well with similar experiments in the human (5).

Desmosomes, hemidesmosomes, and tonofilaments were the structures primarily examined in this study because there is evidence that these organelles give the buccal tissue some physical support by holding it together as an integral unit. Evidence to support this concept comes from the work of Charles and Smiddy (14) who studied the hemidesmosomes and tonofilaments in human epidermis. It was discovered that in specimens of stretched skin, the hemidesmosomes appeared to have been pulled inward toward the centers of the basal

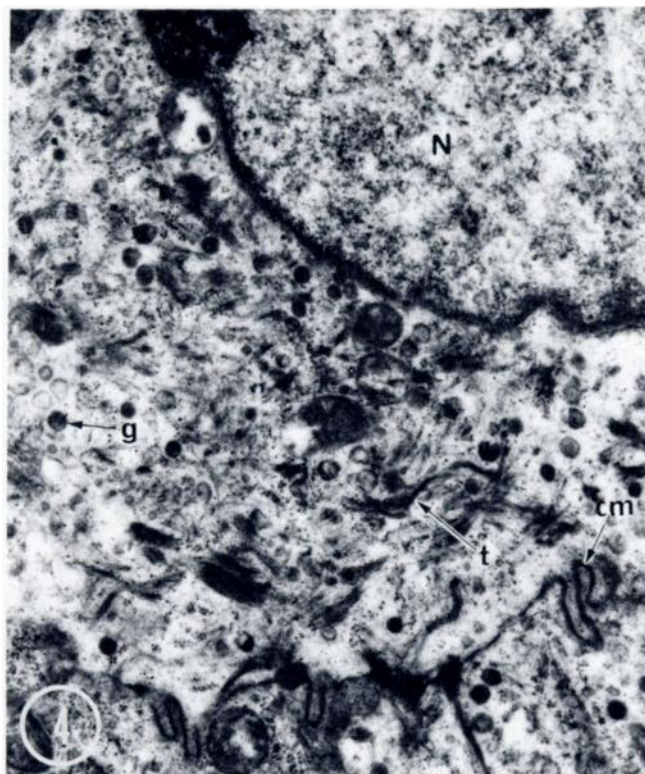


FIG. 4. Typical view of part of a germinal epithelial cell of group II. Note the high concentration of microgranules (g) throughout the cytoplasm. Other organelles include ribosomes, tonofilaments (t), cell membrane (cm) and nucleus (N). $\times 20,000$.

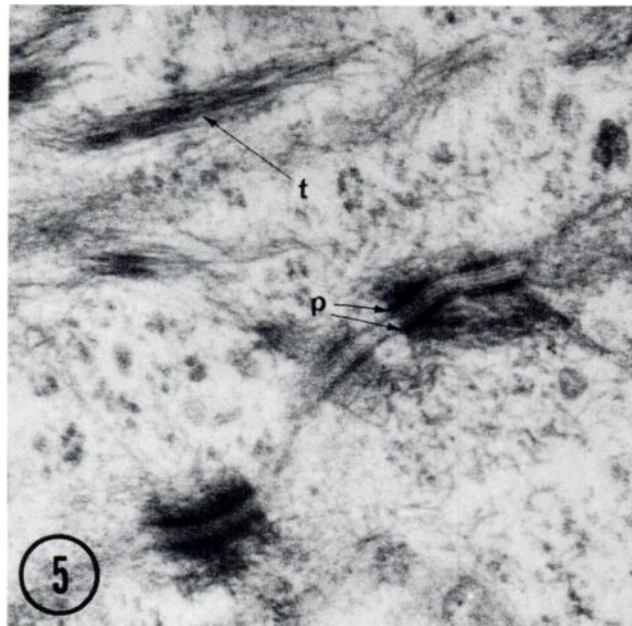


FIG. 5. High magnification of typical desmosomes in group II. The tonofilaments (t) are somewhat indistinct but show little difference in configuration from those in group I (Fig. 3). Only short segments are seen extending from the attachment plaques (p), indicating fragmentation or winding in and out of the plane of sectioning. This view of desmosomes is typical of the material in this group. $\times 75,000$.

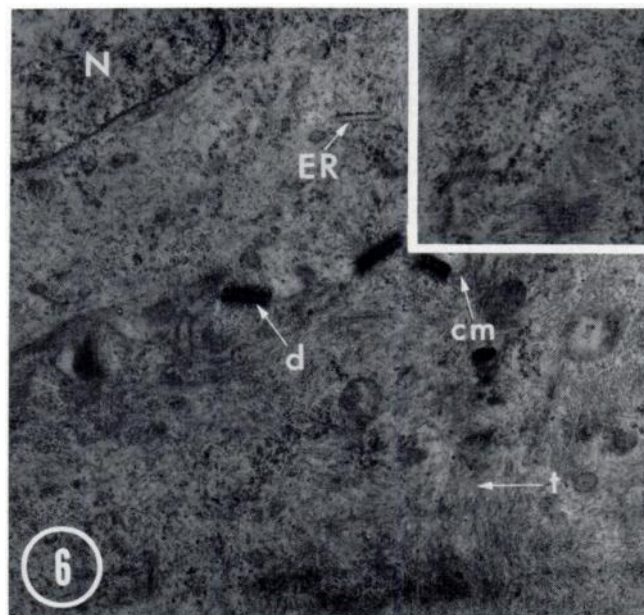


FIG. 6. Typical view of the germinal epithelial cells of group III. Note the greater overall density of tonofilaments (t) than that found in groups I and II. Also present are rough endoplasmic reticulum (ER), desmosomes (d), cell membrane (cm), and nucleus (N). $\times 15,000$. Inset demonstrates the tonofilament bundles and both free and attached ribosomes. Some ribosomes are clustered into polyribosomes. $\times 25,000$.

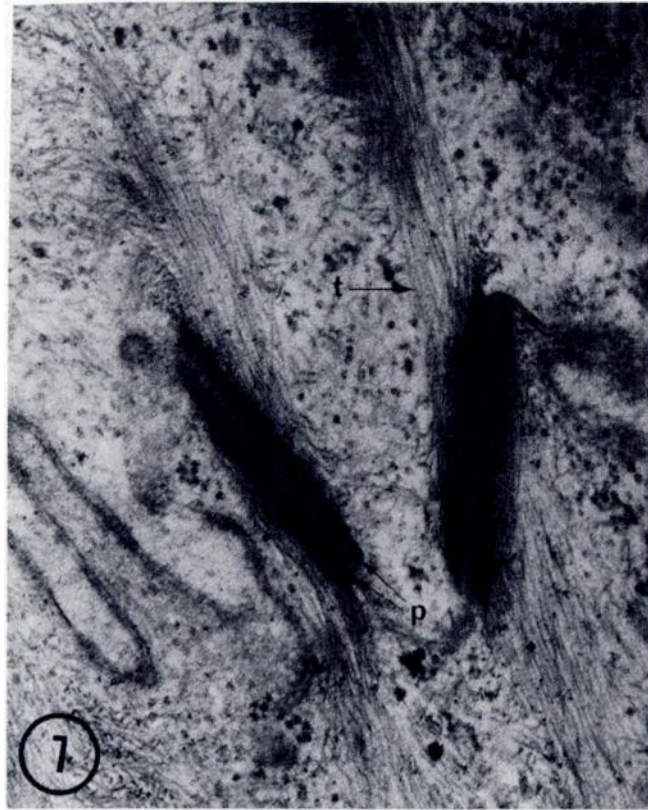


FIG. 7. High magnification of typical desmosomes in group III. The tonofilaments (t) appear to extend as parallel arrays for longer distances away from the attachment plaques (p) in the plane of sectioning than has been noted for groups I and II (Figs. 3 and 5). They may be less fragmentary or follow a more direct pathway from the plaques than those of the other two groups. $\times 60,000$.

TABLE 2
Summary of results

Parameters	Group averages		
	I	II	III
Width of desmosomes, mean	58.8 nm	68.5 nm	65.9 nm
Basal lamina	No differences noted between groups		
Tonofilaments	Random, appear as short segments	Similar appearance as in group I	More densely distributed, fairly long segments
Ribosomes	Free	Some polysomes	Numerous polysomes
Microgranules	Few	Many	Few

cells. Stern (15) discusses their results, extends his analysis to include the desmosomes and argues convincingly that the hemidesmosomes hold the epithelium to the connective tissue (lamina propria) while the tonofilament--desmosome system (with attachment plaques of adjoining epithelial cells presumably "cemented" together) serves to distribute stresses throughout the epithelium and hold it together as a unit. According to this model, when the cells are pulled or stretched, the tonofilaments pull on the attachment plaques of the desmosomes and hemidesmosomes to which they are securely fastened. This would be expected to cause the distance between attachment plaques and the width of the lamina lucida and perhaps the lamina densa to become slightly greater than when no tension is applied.

As noted earlier, the distance between attachment plaques of adjacent cells is smaller in the scorbutic group than in groups II and III. Tonofilaments in this group appear as short segments or as though following a tortuous route through the cytoplasm. If tonofilaments distribute stress in a tissue as suggested by Charles and Smiddy (14) they may, when properly connected, exert a constant tension on the attachment plaques of the desmosomes. If, as is possibly the case for the scorbutic group (I), some or many of these connections are absent or if the tonofilaments are distorted, broken, or follow a winding route, the plaques may be allowed to move closer together. This is in fact the case for group I. The desmosomes for group II are of nearly the same width as those in group III (Table I). These data lead one to speculate that the tonofilament-desmosome complexes may be more tensile or intact in groups II and III than those in group I. The numerous groups of straight tonofilaments seen in group III (Fig. 7) supply added evidence for the possibility that there is increased tension on these structures in animals receiving ascorbic acid. One could then theorize that vitamin C has a role in maintaining proper integrity and orientation of the tonofilaments themselves.

The fact that there is no significant difference in desmosome width between groups II and III (Table I) is not completely resolved at this time and requires further comment. Perhaps there is a sufficient number of intact tonofilaments in group II tissues to hold the attachment plaques away from each other at a similar distance to that found in group III while the "better" oriented filaments in the latter group add extra strength, which may not be demonstrable under the conditions (unstressed) of this experiment.

The electron transparent region of the basement lamina (lamina lucida) contains an unidentified homogeneous fluid or gel which, according to Stern (15), "glues" the basal cell plasma membrane to the adjacent tissue. Priest (8) studied mouse parietal yolk sac and found that the synthesis of epithelial basement lamina is dependent on the presence of vitamin C. Pierce et al. (16) report that the basement lamina is produced by the endoplasmic reticulum of the basal cells. They conducted an *in vivo* study using ferritin-labeled antibodies for basement lamina and found them localized in

the endoplasmic reticulum of the basal epithelial cells. In an excellent review of the literature, Frithiof (17) has also revealed evidence that verifies that the basement lamina is of epithelial origin and can be formed without contact with the underlying lamina propria. Stern (15) has reported the presence of minute vesicular invaginations, or caveolae, of the plasma membrane along the basal surface of the basal cells. He suggests that these may be related to vesicular transport and one may speculate that they function in the movement of materials toward the basement lamina, ultimately for its production. The function of the caveolae cannot, however, be theorized with any degree of certainty on the basis of static micrographs. In any case, the results of the present experiment indicate that vitamin C is not necessary for the formation of the caveolae. Because the basement lamina appears to be similar in all three groups, vitamin C does not seem to be necessary for its maintenance during the experimental period of 20 days.

As ascorbic acid is essential in the formation of several extracellular structures that are thought to be involved in the physical integrity of a tissue (e.g., collagen and lamina densa), it is possible that the vitamin also has a previously undetected role in the synthesis of intracellular constituents and/or the interattachment of these structures to form integral units within the cell. One is inclined to speculate that this vitamin is necessary for the formation of tonofilaments or their attachment to desmosomes and hemidesmosomes. A role such as this would explain the differences for tonofilaments observed between groups. The results of this experiment suggest that vitamin C may be advantageous in larger amounts than those recommended for the prevention of scurvy to enhance the integrity and function of the tonofilament-desmosome systems within the buccal epithelial cells.

Silverman (18) has studied electron opaque microgranules similar to those seen in this study. He has compared them in an ultrastructural study of the cytoplasm of basal and parabasal cells from normally and abnormally keratinized human buccal epithelium. They are thought to play a role in the secretion of intercellular substances during the keratinization process. However, the significance of these organelles cannot be interpreted from the data

accumulated in the present study. This phenomenon needs further study with doses of ascorbic acid intermediate to those used in this experiment.

It is not possible to say with assurance what the direct cause is for the enlarged intercellular spaces in group I. One could speculate that they may be due to decreased production of cementing substances between cells, with a resultant separation of adjacent membranes. This possibility is suggested by the evidence that ascorbic acid is involved in the formation of structures which influence the physical integrity of tissues and may therefore be involved in the synthesis of intercellular cementing materials. The spaces could also be the result of a partial degradation or lack of replacement, or both, of some of the previously existing cementing materials due to a lack of vitamin C. Another possible explanation is the presence of an osmotic imbalance in the tissues. If vitamin C is directly or indirectly responsible for the tonicity either within these cells or in extracellular fluids, its absence might alter tonicities sufficiently to cause the inside of the cells to become hypotonic and thereby shrink as water is lost from them into the intercellular spaces. The latter would then widen with the increased volume of extracellular fluid. This idea is, of course, highly theoretical. 

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